S2 Supplementary Data

Complete A1r/A2r contrast tables, A1 decomposition, Pass B/C moderation tables, and descriptive summaries

Article type: Supplement

Short title: S2 Data - RVC_SLEEP

Robin Langell[1]
[1] Langell Konsult AB, Vallentuna, Sweden
For correspondence: Robin Langell, hello@noeticdiffusion.com

S2.0 Complete A1r table (event vs. matched-control), all three datasets

Dataset	Endpoint	N subj.	Delta	perm p	FDR q
ANPHY	m	27	+0.4302	< 0.001	< 0.001
ANPHY	d	27	+0.2026	< 0.001	< 0.001
ANPHY	e	27	+0.0999	< 0.001	< 0.001
ANPHY	d_n	27	−0.1889	< 0.001	< 0.001
ANPHY	d_l	27	+0.3920	< 0.001	< 0.001
ANPHY	d_s	27	+0.5003	< 0.001	< 0.001
ANPHY	e_e	27	+0.0495	0.082	0.082 (ns)
ANPHY	e_s	27	+0.0773	0.0015	0.0016
ANPHY	e_m	27	+0.4876	< 0.001	< 0.001
BOAS	m	112	+0.2296	< 0.001	< 0.001
BOAS	d	112	+0.4035	< 0.001	< 0.001
BOAS	e	112	+0.2740	< 0.001	< 0.001
BOAS	d_n	112	+0.5893	< 0.001	< 0.001
BOAS	d_l	112	+0.3489	< 0.001	< 0.001
BOAS	d_s	112	+0.4272	< 0.001	< 0.001
BOAS	e_e	112	+0.1030	0.0015	0.0016
BOAS	e_s	112	+0.4851	< 0.001	< 0.001
BOAS	e_m	112	+0.3967	< 0.001	< 0.001
RichSleep	m	24	+0.2296	< 0.001	0.0006
RichSleep	d	24	+0.3487	< 0.001	0.0006
RichSleep	e	24	+0.0190	0.139	0.139 (ns)
RichSleep	d_n	24	+0.2918	< 0.001	0.0006
RichSleep	d_l	24	+0.4330	< 0.001	0.0006
RichSleep	d_s	24	+0.1480	< 0.001	0.0006
RichSleep	e_e	24	−0.2216	< 0.001	0.0006
RichSleep	e_s	24	+0.2229	< 0.001	0.0006
RichSleep	e_m	24	+0.5564	< 0.001	0.0006

Table 1: Complete Pass A1r table: subject-weighted event-vs-matched-control delta, sign-flip permutation p, BH-FDR q (ANPHY/BOAS: 18-test family; RichSleep: separate 9-test family, R6). RichSleep figures are from the QA-corrected `neuralmanifolddynamics_RichSleep_20260709_100219` ingest (S1.0). Sources: `results/rvc_sleep_20260703/02c_pass_a1r_subject_weighted_specificity/pass_a1r_subject_weighted_specificity.csv`; `results/rvc_sleep_20260709/09_richsleep_verification/richsleep_a1r_specificity.csv`.

S2.1 Complete A2r table (event vs. pre_far, within-spindle timing), all three datasets

Dataset	Endpoint	N subj.	Delta	perm p	FDR q
ANPHY	m	27	-0.0131	0.650	0.731 (ns)
ANPHY	d	27	$+0.0402$	0.012	0.037
ANPHY	e	27	-0.0099	0.491	0.632 (ns)
ANPHY	d_n	27	$+0.1056$	< 0.001	0.009
ANPHY	d_l	27	$+0.0081$	0.719	0.761 (ns)
ANPHY	d_s	27	$+0.0177$	0.644	0.731 (ns)
ANPHY	e_e	27	-0.0637	< 0.001	0.009
ANPHY	e_s	27	$+0.0617$	0.024	0.062 (ns)
ANPHY	e_m	27	$+0.0356$	0.119	0.215 (ns)
BOAS	m	110	-0.0410	0.088	0.196 (ns)
BOAS	d	110	$+0.0721$	0.004	0.014
BOAS	e	110	$+0.0368$	0.098	0.196 (ns)
BOAS	d_n	110	$+0.1651$	0.002	0.011
BOAS	d_l	110	$+0.0185$	0.419	0.581 (ns)
BOAS	d_s	110	$+0.0632$	0.139	0.228 (ns)
BOAS	e_e	110	-0.0186	0.381	0.571 (ns)
BOAS	e_s	110	$+0.1141$	0.002	0.011
BOAS	e_m	110	$+0.0053$	0.814	0.814 (ns)
RichSleep	m	24	$+0.0766$	0.325	0.585 (ns)
RichSleep	d	24	$+0.0649$	0.219	0.494 (ns)
RichSleep	e	24	-0.0340	0.397	0.595 (ns)
RichSleep	d_n	24	$+0.0470$	0.720	0.720 (ns)
RichSleep	d_l	24	$+0.0814$	0.048	0.144 (ns)
RichSleep	d_s	24	$+0.0485$	0.642	0.720 (ns)
RichSleep	e_e	24	-0.1005	< 0.001	0.0045
RichSleep	e_s	24	$+0.0477$	0.551	0.709 (ns)
RichSleep	e_m	24	$+0.1594$	0.008	0.036

Table 2: Complete Pass A2r table: subject-balanced event-vs-pre_far within-spindle delta, sign-flip permutation p, BH-FDR q. RichSleep figures are from the QA-corrected `neuralmanifolddynamics_RichSleep_20260709_100219` ingest, $N = 24$ subjects, 160 paired events. The two headline endpoints (`d_n`, `e_s`) remain non-significant at this event count relative to ANPHY (1,037) or BOAS (575); `e_e` and `e_m` newly reach FDR significance and are reported but are not the endpoints the manuscript leads with (R6). ANPHY d_n t-based 95% CI includes zero. Sources: `results/rvc_sleep_20260703/03b_pass_a2r_subject_balanced_profile/pass_a2r_subject_balanced_profile.csv`; `results/rvc_sleep_20260709/09_richsleep_verification/richsleep_a2r_profile.csv`.

S2.2 A1 decomposition (full 9-endpoint family, ANPHY and BOAS)

Endpoint	ANPHY A / B / C	BOAS A / B / C
m	$-0.013/+0.442/+0.429$	$-0.041/+0.295/+0.254$
d	$+0.040/+0.162/+0.202$	$+0.074/+0.340/+0.414$
e	$-0.010/+0.110/+0.101$	$+0.039/+0.245/+0.283$
d_n	$+0.106/-0.293/-0.187$	$+0.166/+0.427/+0.593$
d_l	$+0.008/+0.382/+0.390$	$+0.020/+0.349/+0.370$
d_s	$+0.018/+0.484/+0.501$	$+0.066/+0.354/+0.420$
e_e	$-0.064/+0.113/+0.050$	$-0.017/+0.134/+0.117$
e_s	$+0.062/+0.017/+0.079$	$+0.116/+0.368/+0.484$
e_m	$+0.036/+0.450/+0.486$	$+0.005/+0.417/+0.422$

Table 3: A = timing component (event − pre_far, subject-mean delta), B = pre-spindle-baseline-vs.-matched-control separation, $C = A + B$ = A1r event-vs-control delta. All values subject-mean deltas (units of the standardized endpoint); statistical tests for each component are in the source CSV. Source: `results/rvc_sleep_20260703/02d_pass_a1_decomposition/pass_a1_decomposition.csv` (54 rows: 9 endpoints × 2 datasets × 3 components).

S2.3 Pass B surface sensitivity (full 36-test table, condensed)

28 of 36 (delta_mode × dataset × endpoint) rows have `surface_delta = 0.000000` exactly (`m, d, d_n, d_l, d_s, e_e, e_s`, both delta modes, both datasets) - the anchor-augmented surface is structurally identical to the primary surface for every endpoint that does not algebraically depend on `e_m`. The remaining 8 rows (all involving `e` or `e_m`) are shown in full; none change FDR status.

Delta mode	Dataset	Endp.	Primary	Anchor-aug	Surf. Δ	Corr.	FDR flip
event_control	ANPHY	e	0.0999	0.0939	−0.0060	0.997	No
event_control	BOAS	e	0.2740	0.2724	−0.0016	0.9996	No
event_control	ANPHY	e_m	0.4876	0.4765	−0.0111	0.358	No
event_control	BOAS	e_m	0.3967	0.3502	−0.0465	0.742	No
event_prefar	ANPHY	e	−0.0099	−0.0106	−0.0008	0.9996	No
event_prefar	BOAS	e	0.0368	0.0358	−0.0009	0.9993	No
event_prefar	ANPHY	e_m	0.0356	−0.0021	−0.0377	0.215	No
event_prefar	BOAS	e_m	0.0053	−0.0409	−0.0461	0.587	No

Table 4: Non-identical rows of the full Pass B table. All 36 rows (including the 28 identical ones) are in the source CSV. Source: `results/rvc_sleep_20260703/04_pass_b_surface_sensitivity/pass_b_surface_sensitivity.csv`.

S2.4 Pass C EAP moderation (full 54-test family, ANPHY and BOAS)

Delta mode	Dataset	Anchor (primary)	Endp.	coef	p	FDR q
event_control	ANPHY	ecg_hrv_rmssd_ms	m	0.00082	0.074	0.132
event_control	ANPHY	ecg_hrv_rmssd_ms	d	0.00055	0.045	0.101
event_control	ANPHY	ecg_hrv_rmssd_ms	e	0.00039	0.033	0.100
event_control	ANPHY	ecg_hrv_rmssd_ms	d_n	0.00007	0.900	0.904
event_control	ANPHY	ecg_hrv_rmssd_ms	d_l	0.00097	< 0.001	0.0015 †
event_control	ANPHY	ecg_hrv_rmssd_ms	d_s	0.00007	0.904	0.904
event_control	ANPHY	ecg_hrv_rmssd_ms	e_e	0.00069	0.097	0.145
event_control	ANPHY	ecg_hrv_rmssd_ms	e_s	0.00014	0.741	0.904
event_control	ANPHY	ecg_hrv_rmssd_ms	e_m	0.00092	0.031	0.100
event_control	BOAS	resp_regular_index	m	−0.667	0.296	0.381
event_control	BOAS	resp_regular_index	d	−0.879	0.093	0.261
event_control	BOAS	resp_regular_index	e	−0.681	0.126	0.261
event_control	BOAS	resp_regular_index	d_n	−1.251	0.145	0.261
event_control	BOAS	resp_regular_index	d_l	−0.926	0.077	0.261
event_control	BOAS	resp_regular_index	d_s	−0.281	0.703	0.703
event_control	BOAS	resp_regular_index	e_e	−0.554	0.369	0.415
event_control	BOAS	resp_regular_index	e_s	−0.722	0.238	0.357
event_control	BOAS	resp_regular_index	e_m	−1.055	0.057	0.261

Table 5: Pass C primary-anchor rows (`event_control` delta mode; the `event_prefar` delta mode and the BOAS `anchor_index_v02` sensitivity-anchor rows follow the identical pattern - all null except the flagged exception). † = ANPHY `d_l` survives pooled FDR ($q = 0.0015$) but fails its own subject-level-slope sensitivity check ($p_{\text{slope, perm}} = 0.140$, ns; see manuscript R5) and is treated as a non-robust pooled-model artifact. All 54 rows (2 delta modes × 3 anchor models × 9 endpoints) are in the source CSV. Source: `results/rvc_sleep_20260703/05_pass_c_eap_moderation/pass_c_eap_moderation.csv`.

S2.5 RichSleep EAP moderation (full anchor family, `event_control` delta mode)

Anchor	Endp.	coef	p	FDR q
`cardioresp_anchor_index` (primary)	`m`	0.0035	0.916	0.916
`cardioresp_anchor_index` (primary)	`d`	0.0235	0.477	0.858
`cardioresp_anchor_index` (primary)	`e`	0.0490	0.150	0.594
`cardioresp_anchor_index` (primary)	`d_n`	−0.0219	0.665	0.899
`cardioresp_anchor_index` (primary)	`d_l`	0.0417	0.198	0.594
`cardioresp_anchor_index` (primary)	`d_s`	0.0423	0.398	0.858
`cardioresp_anchor_index` (primary)	`e_e`	0.0765	0.035	0.316
`cardioresp_anchor_index` (primary)	`e_s`	0.0108	0.799	0.899
`cardioresp_anchor_index` (primary)	`e_m`	−0.0147	0.709	0.899
`anchor_index` (sensitivity)	`e_e`	-1.5×10^{-8}	< 0.001	0.0038 †
`anchor_index` (sensitivity)	`e_m`	2.0×10^{-8}	< 0.001	< 0.001 †
`anchor_index_v02` (sensitivity)	`d`	-9.7×10^{-9}	< 0.001	< 0.001 †
`anchor_index_v02` (sensitivity)	`d_l`	-1.1×10^{-8}	< 0.001	< 0.001 †
`sympathetic_index` (sensitivity)	`e_e`	-7.5×10^{-9}	< 0.001	0.0038 †
`sympathetic_index` (sensitivity)	`e_m`	1.0×10^{-8}	< 0.001	< 0.001 †
`resp_anchor_index` (sensitivity)	`m`	0.0524	0.227	0.396
`resp_anchor_index` (sensitivity)	`d`	0.0061	0.932	0.932
`resp_anchor_index` (sensitivity)	`d_l`	0.0387	0.468	0.596
`resp_anchor_index` (sensitivity)	`e_m`	−0.0099	0.891	0.920
`ocular_stability_index` (sensitivity)	`d_l`	-2.2×10^{-9}	< 0.001	< 0.001 †

Table 6: RichSleep EAP-moderation table, `event_control` delta mode, $N = 24$ subjects, on the QA-corrected `neuralmanifolddynamics_RichSleep_20260709_100219` export (the `event_prefar` mode and all coefficients/SEs are in the source CSV, 126 total rows across both delta modes and 7 anchors). † = nominal FDR survivor with a coefficient magnitude ($\approx$ 10^{-8}-10^{-9} per unit of a large-scale raw composite index) read as a scale/multicollinearity artifact rather than genuine moderation (manuscript R6, S1.7). `resp_anchor_index`'s `event_control`-mode coefficients are a clean null, matching the primary anchor's pattern exactly - see S2.5a for an exploratory pattern found under an alternative delta-mode specification. The primary, mechanistically-motivated anchor (`cardioresp_anchor_index`) shows no FDR survivors at all. Source: `results/rvc_sleep_20260709/09_richsleep_verification/richsleep_eap_moderation.csv`.

S2.5a `resp_anchor_index`: sensitivity-anchor exploratory result

The `resp_anchor_index` sensitivity model in Table Table 6 is null in the primary `event_control` delta mode ($p \approx 0.23$-0.93 on `m`, `d`, `d_l`, `e_m`), matching the primary anchor's null pattern. Under the alternative `event_prefar` delta-mode specification, a weaker, exploratory pattern emerges on four endpoints:

Endp.	coef	p	FDR q	joint q (108)	excl. slope-outlier	excl. value-outlier
`d`	$+0.111$	0.0087	0.0151	0.0213	$p = 0.075$ (loses sig. by Wilcoxon)	$p = 0.0077$ (holds)
`e`	$+0.102$	0.0003	0.0006	0.0010	$p = 0.005$ (strengthens)	$p = 0.0002$ (holds)
`d_n`	$+0.285$	0.0005	0.0011	0.0017	$p = 0.02$ (holds)	$p = 0.0005$ (holds)
`e_s`	$+0.205$	0.0002	0.0006	0.0009	$p = 0.009$ (holds)	$p = 0.0002$ (holds)
`e_e`	$+0.030$	0.035	0.0501 (boundary)	0.0732 (fails)	-	$p = 0.039$ (unchanged, still weak)

Table 7: `resp_anchor_index_pre_far` moderation, `event_prefar` delta mode, with two robustness checks beyond scale/multicollinearity/single-subject-outlier (all of which this pattern also passes: $\approx [-1, 1]$ anchor scale, max $|r| = 0.18$ with the primary anchor/covariates). **Joint q (108)**: BH-FDR recomputed pooling both delta modes' 108 sensitivity-anchor tests into one family - a stricter check against testing two delta-mode specifications; `d`/`e`/`d_n`/`e_s` all survive ($q \leq 0.021$), `e_e` does not ($q = 0.073$) and is downgraded from boundary to non-significant. **Excl. slope-outlier**: Wilcoxon signed-rank p on subject-level slopes after excluding the single subject with the most extreme fitted slope; `d` alone loses significance. **Excl. value-outlier**: WLS-cluster coefficient p after excluding the single subject with the most extreme **raw** `resp_anchor_index` value instead - a different, complementary outlier check; all coefficients are essentially unchanged, none lose significance. Source: `results/rvc_sleep_20260709/11_richsleep_resp_anchor_followup/{resp_anchor_robustness_checks,check5_joint_fdr_108,check6_anchor_value_outlier_exclusion}.csv`.

This pattern is not a scale, multicollinearity, or single-outlier artifact (in either the slope-influence or raw-value sense), and holds up under a substantially stricter joint-family FDR correction across both delta-mode specifications tested. Three of the four survivors (`e`, `d_n`, `e_s`) pass every check attempted; `d` is the one exception, losing significance specifically when the most-influential-slope subject is excluded (though not when the most-extreme-value subject is excluded instead) - a genuine, if narrower, weakness worth naming rather than glossing over. `e_e`, previously reported at the FDR boundary, is downgraded to non-significant under the stricter joint correction and should not be read as a fifth survivor. This anchor's `event_control`-mode signal was previously found to be null after an upstream construction-bug fix (manuscript R6); given that history, "passed every artifact check attempted" is treated as a hypothesis for a dedicated, independently-replicated test - not as proof of a genuine effect.

S2.6 RichSleep phase-resolved EAP mechanism test (Aim B + Aim C, manuscript R6b)

Level	Channel	N	R	p
Within-subject, median across 24 subj.	`phi_cardiac_onset`	24 subj.	0.307 (median)	0/24 FDR-sig.
Within-subject, median across 24 subj.	`phi_resp_onset`	24 subj.	0.324 (median)	0/24 FDR-sig.
Second-order (population)	`phi_cardiac_onset`	24 subj.	0.160	0.547
Second-order (population)	`phi_resp_onset`	24 subj.	0.109	0.755

Table 8: Aim B: spindle-onset cardiac/respiratory phase-locking, Rayleigh test (Zar 1999 large-sample approximation). "Within-subject" summarizes 24 independent per-subject Rayleigh tests per channel (full per-subject R/p/FDR-q in the source CSV); "second-order" is a single Rayleigh test on the 24 subject-level mean-direction angles, testing for a common population-level preferred phase. Both null at every level. Subjects with $n =$ 1–2 events are not Rayleigh-testable. Source: `results/rvc_sleep_20260708/10_richsleep_phase_resolved_eap/molB_within_subject_phase_locking.csv`, `molB_second_order_population_phase.csv`.

Delta mode	Endpoint	N events	coef	p	FDR q
`event_control`	`m`	172	-1.03×10^4	0.839	0.904
`event_control`	`d`	172	$+1.06 \times 10^4$	0.658	0.904
`event_control`	`e`	172	-4.77×10^3	0.882	0.904
`event_control`	`d_n`	172	$+5.82 \times 10^3$	0.904	0.904
`event_control`	`d_l`	172	-4.09×10^3	0.903	0.904
`event_control`	`d_s`	172	$+8.56 \times 10^4$	0.100	0.904
`event_control`	`e_e`	172	-4.08×10^4	0.323	0.904
`event_control`	`e_s`	172	$+3.90 \times 10^4$	0.231	0.904
`event_control`	`e_m`	172	$+1.25 \times 10^4$	0.836	0.904
`event_prefar`	`m`	160	$+1.56 \times 10^5$	0.175	0.789
`event_prefar`	`d`	160	-1.50×10^4	0.819	0.922
`event_prefar`	`e`	160	-2.03×10^4	0.673	0.922
`event_prefar`	`d_n`	160	-1.35×10^5	0.531	0.922
`event_prefar`	`d_l`	160	$+4.23 \times 10^4$	0.275	0.826
`event_prefar`	`d_s`	160	$+3.07 \times 10^4$	0.734	0.922
`event_prefar`	`e_e`	160	-9.06×10^2	0.962	0.962
`event_prefar`	`e_s`	160	-5.19×10^4	0.604	0.922
`event_prefar`	`e_m`	160	$+8.64 \times 10^4$	0.158	0.789

Table 9: Aim C: `hep_amplitude_pre_far` moderation of the spindle-locked geometry response, WLS-cluster (weights $= 1/n_{\text{events, subject}}$, cluster-robust SE by subject) + centered nuisance covariates, identical machinery to the manuscript R6/Pass C moderation models. Large coefficient magnitudes ($\approx 10^4$-10^5) reflect `hep_amplitude_pre_far`'s presumably microvolt-scale raw units, not an anomaly. 0/18 tests FDR-significant; companion subject-level-slope sign-flip permutation test (source CSV) agrees at every endpoint. Source: `results/rvc_sleep_20260708/10_richsleep_phase_resolved_eap/molC_hep_moderation.csv`.

Both mechanism-level tests use the window-N2, geometry-valid RichSleep event-bin population (190 of 2,253 raw `event`-bin rows; 24 subjects, 7-8 events/subject on average). The 190/2,253 split is a stage-window mismatch (S1.1, S1.8): the detector used N2 hypnogram epochs; the event-locked `stage` label is the 30 s MNPS window centre. This is a conservative construct-validity filter, not evidence that YASA detected non-N2 spindles. Tonic EAP mixed-bin row counts (2,260–2,435) are a different object and must not be compared with 190 as unique-spindle N.

S2.7 Nuisance-covariate coverage by pass

Pass	Covariates retained ($\geq 50\%$ coverage)
Pass C, ANPHY	`event_duration_sec`, `window_center_sec`, `ocular_stability_index`, `sympathetic_index`, `vagal_index`, `eog_quality_quality`, `ecg_quality_quality`, `anchor_support_fraction_quality`
Pass C, BOAS	same family as ANPHY, coverage-permitting per anchor model
RichSleep EAP moderation	`event_duration_sec`, `window_center_sec`, `ocular_stability_index`, `sympathetic_index`, `vagal_index`, `eog_quality_quality`, `ecg_quality_quality`, `resp_quality_quality`, `cardioresp_quality_quality`, `anchor_support_fraction_quality`

Figure 1: Covariates actually included after `available_covariates()` filtering (S1.6). `frequency_hz` is requested defensively in every pass but is never present in the NMD event-locked schema and is always dropped.

S2.8 Descriptive summary by dataset

Dataset	*N* subj.	Median events/ subj.	Range	Total spindle events
ANPHY	27	≈ 40	-	1,076
BOAS (ds005555)	128	5	1-18	634
RichSleep	24	≈ 161	14-362	46,058 raw (all bins) / 3,900 filtered

Figure 2: Per-subject spindle-event distribution. BOAS's sparse, uneven event counts (median 5, range 1-18) directly motivated the subject-weighted A1r/A2r protocol amendment (S1.6). RichSleep's per-subject counts (14-362, from `event_level_richsleep.parquet`) are comparably uneven; the same subject-weighted machinery was applied without modification. All 24 subjects are usable on this ingest.

Bibliography